

Ahmed Mamdouh

Senior Full-Stack Software Engineer · JavaScript / TypeScript / Node.js / NestJS · PHP / Laravel · React / Next.js / Vue.js

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SUMMARY

Senior full-stack engineer with 10+ years building and shipping high-scale, distributed web systems. Deep expertise across the JavaScript/TypeScript stack — Node.js, NestJS, React, Next.js, and Vue.js — alongside PHP/Laravel for backend services and APIs. Built real-time platforms handling 100k+ concurrent connections, GraphQL services processing millions of daily events, and fintech systems clearing thousands of transactions per day. Strong on system design, REST and GraphQL API architecture, microservices, observability, and cloud infrastructure (AWS, Kubernetes, Docker, CI/CD). Comfortable owning end-to-end delivery from product spec to deployment and on-call, and led teams of 7+ engineers while still writing and reviewing code daily.

PROFESSIONAL EXPERIENCE

Netsync Network Solutions — Senior Software Engineer (Remote, US)

Feb 2023 – Present

IT solutions provider — cloud infrastructure, cybersecurity, and IPTV device management for enterprise clients.

- **ALTO IPTV platform:** designed a real-time device management system on Node.js, NestJS, Socket.io, and MongoDB handling 100,000+ concurrent device connections.
- **GraphQL APIs:** designed and shipped schema-first GraphQL services as the unified API layer for web, mobile, and partner integrations.
- **Microservices architecture:** broke a monolith into NestJS services on Docker/Kubernetes, improving p95 latency by ~40% and isolating failure domains.
- **Real-time event pipeline:** WebSocket + queue-based notification system delivering millions of events daily with at-least-once semantics.
- **DevOps & reliability:** owned CI/CD on Jenkins, Kubernetes deployments on AWS (EKS), monitoring, and incident response.
- **Engineering leadership:** led 7+ engineers across Node.js, NestJS, and Angular; ran code reviews, design docs, and hiring.

Tech Stack: TypeScript, Node.js, NestJS, GraphQL, Socket.io, MongoDB, PostgreSQL, AWS (EKS, S3), Docker, Kubernetes, Jenkins

SweepSouth — Senior Software Engineer (Remote, South Africa)

Apr 2022 – Feb 2023

Africa's largest on-demand home services platform — operating across South Africa, Kenya, Nigeria, and Egypt (FilKhedma).

- **Backend services:** built and scaled Node.js / NestJS services serving 100k+ active customers.
- **Performance:** improved booking-flow throughput and reduced API tail latency through query tuning, caching, and async workflows.
- **Frontend:** shipped customer-facing flows in Angular and Vue.js with a focus on responsive, mobile-first UX.
- **Infra:** containerized services with Docker/Kubernetes on AWS; standardized deploy + rollback workflow.

Tech Stack: TypeScript, Node.js, NestJS, MongoDB, Angular, Vue.js, Docker, Kubernetes, AWS

Technocloud — Tech Lead (Cairo, EG)

Aug 2020 – Apr 2022

Multiple fintech and e-commerce products: Zerocash (e-wallet), Rojetah (healthcare), Dealmart (e-commerce), Trjim.

- **Zerocash e-wallet:** architected a secure transaction system processing thousands of daily payments with audit trails and reconciliation.
- **Rojetah healthcare:** built a patient/provider platform — appointment booking, messaging, and records.
- **API design:** shipped RESTful APIs in Laravel/PHP and Node.js powering web and partner clients with consistent versioning and contract tests.
- **Performance:** ~45% improvement on critical endpoints via caching (Redis), N+1 cleanup, and indexed queries.
- **Team lead:** led 7+ engineers in Scrum; ran sprint planning, estimation, and code reviews.

Tech Stack: PHP, Laravel, Vue.js, Node.js, MySQL, Redis, RESTful APIs, Docker

Itrinity (UptimeRobot) — Senior Software Engineer (Remote, Slovakia)

Oct 2019 – Aug 2020

Website monitoring SaaS — 2.1M+ users, real-time uptime monitoring and alerting at scale.

- **Monitoring services:** built high-throughput Node.js / NestJS services running millions of website checks daily.
- **GraphQL:** designed schema-first GraphQL APIs that reduced client network overhead by ~35%.
- **Event-driven alerting:** queue-based notification pipeline with retry, deduplication, and SLA tracking.
- **AWS:** deployed on ECS and Lambda; managed configuration, secrets, and rollouts.

Tech Stack: TypeScript, Node.js, NestJS, GraphQL, AWS (ECS, Lambda)

Schoolver — Software Engineer (Cairo, EG)

Mar 2017 – Oct 2019

Comprehensive School Management System (SMS) and Learning Management System (LMS).

- **LMS:** built end-to-end with a Vue.js frontend and REST APIs in Laravel/PHP and Node.js serving thousands of students and teachers.
- **Backend services:** Laravel/PHP and Node.js REST services powering web clients with consistent API contracts.
- **Real-time features:** Socket.io chat and notifications.
- **Multi-tenancy:** designed schema and routing to support multiple schools on shared infrastructure.
- **Practices:** introduced CI/CD, code reviews, and automated tests across the team.

Tech Stack: PHP, Laravel, Vue.js, Node.js, MySQL, Socket.io, CI/CD

TECHNICAL SKILLS

Languages: TypeScript, JavaScript (ES2022+), PHP, SQL

Backend: Node.js, NestJS, Express, Laravel, GraphQL (Apollo), Socket.io, REST, Microservices, Event-driven

Frontend: React, Next.js, Vue.js, Nuxt.js, Angular, Redux, Tailwind CSS

Data: PostgreSQL, MongoDB, MySQL, Redis, Prisma ORM, Eloquent ORM

Infra & DevOps: AWS (EKS, ECS, Lambda, S3), Docker, Kubernetes, Jenkins, GitHub Actions, Linux

Architecture: System design, distributed systems, microservices, event-driven, multi-tenancy, RBAC

Testing: Jest, Mocha, Chai, Cypress, PHPUnit; contract + integration testing

Integrations: Stripe, PayPal, Paymob, Paytabs, Twilio, OAuth 2.0, Slack / LinkedIn / Google APIs

NOTABLE PROJECTS

- **Go2alto (Netsync):** IPTV device management at 100k+ concurrent — Node.js, NestJS, GraphQL, MongoDB, Socket.io.
- **UptimeRobot enhancements:** monitoring microservices — Node.js, NestJS, GraphQL, AWS.
- **Zerocash:** secure e-wallet platform — Laravel/PHP, Vue.js, Node.js, MySQL.
- **FilKhedma / SweepSouth:** on-demand home services platform — NestJS, MongoDB, Angular, Vue.js.
- **Mall of Arabia:** Egypt's largest mall website — Next.js / Nuxt 3, Tailwind.
- **Open source:** 44+ public repositories — Node.js, NestJS, TypeScript, and PHP/Laravel projects (github.com/xerk).

EDUCATION

Akhbar El Youm Academy — *B.Sc. Computer Science*

2014 – 2018

Languages: Arabic (native) · English (professional working proficiency)